# FROM ORBIT TO FIELD

How satellite data are obtained, processed and evaluated for agriculture and soil health

## 1 ACQUIRE

Sensors in orbit record energy reflected or emitted by the land surface.

- ▸ Optical multispectral: Sentinel-2 (10-20 m, 5-day), Landsat 8/9 (30 m, 8-day), MODIS (250 m, daily)
- ▸ Hyperspectral: EnMAP, PRISMA, EMIT (hundreds of narrow bands)
- ▸ Radar (SAR): Sentinel-1 C-band; works through cloud, sensitive to moisture & structure
- ▸ Thermal: Landsat TIRS, ECOSTRESS for crop water stress / ET
- ▸ Passive microwave: SMAP, SMOS for coarse soil moisture

## 2 PRE-PROCESS

Turn raw digital numbers into physically meaningful, co-registered surface data.

- ▸ Radiometric calibration -> top-of-atmosphere radiance / reflectance
- ▸ Atmospheric correction (Sen2Cor, LaSRC, 6S) -> surface reflectance (Level-2A)
- ▸ Cloud, shadow and snow masking (s2cloudless, Fmask, SCL)
- ▸ Orthorectification and co-registration to a map grid; harmonisation across sensors (e.g., HLS)
- ▸ Temporal compositing, gap-filling and smoothing of time series

## 3 DERIVE FEATURES

Compress spectra into indices and biophysical variables that track crops and soil.

- ▸ Vegetation indices: NDVI, EVI, NDRE, SAVI, GNDVI; red-edge chlorophyll / nitrogen proxies
- ▸ Biophysical retrieval: LAI, fAPAR, canopy chlorophyll via PROSAIL inversion or ML
- ▸ Bare-soil composites (multi-year, NDVI < ~0.25) expose topsoil spectra for SOC, clay, CaCO3, pH, CEC
- ▸ Soil & water indices: NDWI, NDMI, brightness, SWIR ratios; SAR backscatter and coherence
- ▸ Phenology metrics: green-up, peak, senescence dates, integrated NDVI

## 4 MODEL

Link features to ground truth with statistical, ML or process-based models.

- ▸ Crop type & cropland mapping: random forest, gradient boosting, CNN / transformer on time series
- ▸ Yield & biomass: regression, LightGBM, LSTM, crop-model data assimilation (WOFOST, DSSAT, APSIM)
- ▸ Soil properties: PLSR / Cubist / RF on bare-soil spectra + terrain, climate & parent-material covariates
- ▸ Microbial proxies (emerging): NDVI/hyperspectral linked to eDNA & PLFA fungal:bacterial data via GPR / RF
- ▸ Digital soil mapping at scale: SoilGrids (250 m) from ~240k profiles and 400+ covariates

## 5 EVALUATE & DELIVER

Quantify accuracy and uncertainty, then translate maps into decisions.

- ▸ Spatial / blocked cross-validation; independent hold-out fields
- ▸ Regression metrics: R2, RMSE, bias, RPD / RPIQ, prediction intervals (e.g., 5-95% quantiles)
- ▸ Classification metrics: overall accuracy, F1, kappa, confusion matrix; area estimation with design-based sampling
- ▸ Transferability checks across years, regions and sensors; sensitivity to cloud gaps
- ▸ Outputs: variable-rate prescription maps, yield forecasts, SOC baselines & MRV, drought early warning

### WHAT CAN BE SEEN FROM SPACE

- Crop type, extent and field boundaries
- Canopy greenness, LAI, chlorophyll, N status
- Phenology, growth anomalies, lodging, flooding
- Crop water stress (thermal) and ET
- Soil moisture (SAR / passive microwave)
- Topsoil SOC, clay, CaCO3, pH & CEC on bare soil
- Tillage, residue cover, cover-crop adoption

### KEY LIMITS & CAVEATS

- Optical sensors are blind under cloud; SAR fills gaps
- Soil signal only from exposed, dry, smooth bare soil
- Satellite SOC/pH/CEC models: typically R2 ~0.4-0.7; RPD >1.4 = fair
- Microbial biomass / F:B ratio is NOT directly observed - only inferred
- Spatial resolution vs field size; mixed pixels
- Models rarely transfer across regions without recalibration
- Ground truth remains essential

### OPEN DATA & TOOLS

- Copernicus Data Space (Sentinel-1/2/3)
- USGS EarthExplorer / NASA Earthdata (Landsat, MODIS, HLS, SMAP)
- Google Earth Engine; Microsoft Planetary Computer
- SoilGrids (ISRIC); LUCAS topsoil (ESDAC); SSURGO (USA)
- USDA NASS Cropland Data Layer; ESA WorldCereal
- FAO WaPOR, GEOGLAM Crop Monitor
- Open software: SNAP, QGIS, GDAL, rasterio, scikit-learn

Sources: Weiss, Jacob & Duveiller (2020) Remote Sens. Environ. 236:111402; Angelopoulou et al. (2019) Remote Sens. 11:676; Lima et al. (2025) Remote Sens. 17:882; Poggio et al. (2021) SOIL 7:217; Gorelick et al. (2017) Remote Sens. Environ. 202:18; Cardille et al. (eds., 2024) Cloud-Based Remote Sensing with Google Earth Engine (open access, Springer).